# 2025 WSMO Speed Round

SMO Team

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**SR 1:** How many permutations $(a_1, a_2, a_3, a_4, a_5)$ of $(1, 2, 3, 4, 5)$ satisfy

$$a_1 + a_2 = a_4 + a_5?$$

**SR 2:** Charlie has five consecutive integers and chooses one of them. The sum of her four numbers that she did not choose is 2025. What number did she choose?

**SR 3:** There exists a unique positive integer $n$ such that $n^3 + 6n^2 + 11n + 21$ is a perfect cube. Find the value of $n$.

**SR 4:** Suppose that three fair six-sided dice are tossed so that the three numbers that they land on are the roots of the monic cubic $P(x) = x^3 + ax^2 + bx + c$. What is the expected value of $200 + 4a + 2b + 4c$?

**SR 5:** Real numbers $a$ and $b$ satisfy

$$\frac{a}{b} + \frac{b}{a} = \frac{a}{b^2} + \frac{b}{a^2} = 10.$$

Then, $ab$ can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**SR 6:** Suppose $\mathcal{R}_1$ and $\mathcal{R}_2$ are rectangles such that every vertex of $\mathcal{R}_1$ lies on a different side of $\mathcal{R}_2$. The side lengths of $\mathcal{R}_1$ are in a $3 : 4$ ratio, and the side lengths of $\mathcal{R}_2$ are in an $8 : 9$ ratio. Then, the ratio between the perimeters of $\mathcal{R}_1$ and $\mathcal{R}_2$ is $m : n$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**SR 7:** Let $ABCDEF$ be an equiangular hexagon with $AB = DE = 4$, $BC = EF = 7$, and $CD = FA$. Point $P$ lies in the interior of the hexagon such that $[APB] = 12\sqrt{3}$, $[BPC] = 14\sqrt{3}$, and $[CPD] = 9\sqrt{3}$. Find the perimeter of the hexagon.

**SR 8:** Alice rolls a number $t$ on a fair six-sided die. Bob then rolls $t$ fair twenty-sided dice, getting $a_1, a_2, \ldots, a_t$. Let $a$ be the maximum integer such that $2^a$ divides $\prod_{i=1}^{t} a_i!$. The expected value of $a$ can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**SR 9:** Let $\mathcal{S}$ denote the set of all permutations of the first 6 positive integers. A permutation $(a_1, a_2, \ldots, a_6)$ is chosen uniformly at random from the elements of $\mathcal{S}$. Suppose that there are $r$ distinct values among the remainders when $a_1 + 1, a_2 + 2, \ldots, a_6 + 6$ are each divided by 6. The expected value of $r$ can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**SR 10:** Complex numbers $a, b, c$ satisfy the following system of equations:

$$\begin{aligned}(a+b)^2 &= ab + 39,\\ (b+c)^2 &= bc + 61,\\ ab + bc + ca &= 38.\end{aligned}$$

Find the sum of all possible values of $(a+b+c)^2$.